

CSE221: Algorithms

Lab-05

Fall 2025

Assignment-05 Links:

codeforces Invitation Link:

<https://codeforces.com/contestInvitation/89feb126a9191ae393d028f5c22232508072d137>

Alternative Link: <https://codeforces.com/contests/650807>

[The alternative link will **only work** after you have successfully registered using the first link.]

Deadline for Assignment-05: Dec 7, 2025 2:00 PM

No submission can be considered after the deadline ends.

Important Notes:

Since the midterm date has been shifted, there is a change in the lab plan as follows:

- **Deadline for Assignment-04 Extended Till:** Dec 7, 2025 2:00 PM
- **Quiz-04 Date:** Dec 6, 2025, Dec 7, 2025 (as per your routine)
- **Quiz-04 Syllabus:** Assignment-04 and Assignment-05 (one quiz covering both topics)

Learning Outcome:

Upon successful completion of this lab, students will have a solid understanding of graph and grid traversal techniques. They will be able to apply fundamental concepts such as graph traversal, cycle detection, shortest path finding, and flood fill in practical problem-solving scenarios. Students will also learn how similar ideas extend to two-dimensional grids, enabling them to identify reachable regions and solve practical search problems. Overall, this lab aims to help students connect the theory of graph algorithms with their practical

use in code, and to develop confidence in choosing and applying appropriate traversal strategies.